

JITHENDRA PATHIRAJA

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Moratuwa, Sri Lanka

EDUCATION

University of Moratuwa <i>B.Sc. Eng. (Hons.), Mechanical Engineering – First Class; 3.73/4.00</i> Dean's List: Semesters 2, 4 and 8	Moratuwa, Sri Lanka 2022 – 2026
Maliyadeva College <i>GCE Advanced Level – Physical Science Stream; Z-score: 2.1052</i> GCE Ordinary Level (2017): 9A Passes	Kurunegala, Sri Lanka 2012 – 2020
Wadugedara Primary College <i>Primary Education</i> Scholarship Examination: 184/200	Kuliapitiya, Sri Lanka 2007 – 2011

EXPERIENCE

Temporary Instructor <i>Department of Mechanical Engineering</i>	Jul 2026 – Present University of Moratuwa
Steering Systems Engineer <i>Team FalconE Racing</i>	May 2025 – Jun 2026 University of Moratuwa
Team Member – Steering <i>Team FalconE Racing</i>	Aug 2024 – May 2025 University of Moratuwa
Trainee Mechanical Engineer <i>Vega Innovations (Pvt) Ltd</i>	Dec 2024 – Jun 2025 Colombo, Sri Lanka

RESEARCH INTERESTS

Computational Fluid Dynamics (CFD), Computational Mechanics, Fluid Mechanics, Aerodynamics, Aerospace Structures, Mechanical Design, Design Optimization, Finite Element Analysis (FEA), Numerical Simulation

ENGINEERING PROJECTS

Reusable Welding Jig for the Vega Street Bike Chassis

- Designed a reusable welding jig to ensure precise alignment and positioning of chassis members during fabrication, minimizing dimensional errors and improving repeatability.
- Manufactured using sheet metal and box section bars; components cut, drilled, and welded per SolidWorks manufacturing drawings.
- Aligned jig using a laser level across all axes prior to chassis assembly to ensure dimensional accuracy.

Rear Paddock Stand for the Vega Street Bike

- Designed and developed a rear paddock stand to elevate the motorcycle's rear wheel for maintenance operations, modeled as a full 3D CAD assembly in SolidWorks.
- Performed FEA using SolidWorks Simulation to verify structural integrity under applied hand force and motorcycle rear weight loading.
- Fabricated using mild steel hollow sections with MIG welding, positioned using a custom welding jig; finished with primer and paint for corrosion protection.

Front Suspension Dynamic Analysis – Vega Street Bike

- Developed a simplified dynamic model of the Vega Street Bike in MATLAB/Simulink to analyze front suspension response under bump excitation.
- Evaluated chassis displacement and studied the effect of suspension stiffness and damping parameters on ride behavior.

- Investigated critical damping characteristics to understand the influence of suspension tuning on vehicle stability and ride comfort.

Design of High-Speed Ambulance Craft

- Designed a high-speed ambulance vessel using regression-based analysis of existing craft data.
- Established relationships between principal dimensions to ensure stability and hydrodynamic performance.
- Selected optimal vessel dimensions based on statistical modeling and design constraints.

Design and Development of a Hydrodynamic Tunnel for Experimental Fluid Dynamics

- Designed and developed a laboratory-scale hydrodynamic tunnel to generate controlled, low-turbulence flow for experimental fluid dynamics studies.
- Performed transient CFD simulations using ANSYS Fluent (SST $k-\omega$ model) to analyze flow behavior, pressure distribution, and optimize flow uniformity.
- Developed a comprehensive design framework for laboratory-scale hydrodynamic tunnels, including component design, engineering calculations, CAD modeling, and design optimization, followed by experimental validation.

Steering System Optimization – Falcon E2 Formula Student Car

- Designed and developed an improved steering mount to enhance rigidity under high steering loads.
- Developed a custom jig for precise alignment of bearing housings during fabrication.
- Integrated and optimized steering system to improve performance and reliability.

Quick Jack System – Falcon E2 Formula Student Car

- Designed and fabricated a mechanical quick jack system for efficient lifting of the vehicle.
- Selected structural members to ensure strength, stability, and safe operation under load.
- Optimized design for ease of use and reliable load distribution.

Prosthetic Leg Fatigue Testing Mechanism

- Designed a mechanical system to simulate human gait for fatigue testing of prosthetic legs.
- Developed linkage mechanism and loading system to replicate realistic walking conditions.
- Modeled and fabricated prototype using SolidWorks and fabrication processes.

Numerical Analysis of NACA 4312 Airfoil

- Performed CFD analysis using ANSYS Fluent to evaluate lift, drag, and pressure distribution.
- Analyzed velocity contours to study aerodynamic performance under varying conditions.

Gearbox Design for Friction Washer

- Designed gearbox system and performed torque and power transmission calculations.
- Modeled assembly components using Solid Edge.

Reverse Engineering of Bicycle Air Pump

- Reverse engineered a bicycle air pump and developed CAD models of internal components.
- Conducted material identification using FTIR and microscopic analysis.

Smart Waste Management System

- Developed a CNN-based waste classification system using TensorFlow and OpenCV.
- Implemented automated sorting using Raspberry Pi and sensor integration.

SKILLS

Languages: English (professional proficiency), Sinhala (native proficiency)

Software:

- **3D Modeling:** SolidWorks, Solid Edge, AutoCAD, Rhinoceros 3D
- **Simulation:** SolidWorks Simulation, ANSYS, MATLAB (Simulink, Simscape Multibody), Orca3D, Maxsurf

Engineering: Mechanical Design, Design for Manufacturing (DFM), CAD Modeling, CFD Analysis, Structural Analysis, Mechanism Design, Steering System Design, Fabrication and Prototyping

Programming: Python, C++

Soft Skills: Communication, Leadership, Teamwork

HONORS & AWARDS

IMechE SL Group Final Year Project Award

Institution of Mechanical Engineers (IMechE), Sri Lanka Group

Dec 2025

- Awarded for outstanding performance, innovation, and teamwork in final year engineering project.
- Selected among top projects evaluated by IMechE Sri Lanka Group.

PROFESSIONAL MEMBERSHIPS

Institution of Mechanical Engineers (IMechE)

Affiliate Student Member | User Number: 80652142

Institution of Engineers Sri Lanka (IESL)

Student Member | Membership Number: S-30322

REFERENCES

Dr. Indrajith D. Nissanka

Senior Lecturer, Department of Mechanical Engineering

Faculty of Engineering, University of Moratuwa

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Dr. Lihil Subasinghe

Senior Lecturer, Department of Mechanical Engineering

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Mr. Chanaka Rathnayaka

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